

Kubernetes Terminal Log Summary

Terminal Screenshots Analysis & Command Execution Details

IMAGE 1

Initial kubectl Apply Errors & Kind Installation

1. Initial Cluster Connection & YAML Validation Failures

When executing `kubectl apply -f` commands, connection and syntax validation errors occurred:

- **Connection Refused Error:** The connection to the server localhost:8080 was refused - Indicates the cluster was not set up or the control plane was not running.
- **YAML Validation Errors:** OpenAPI validation failed across sub-files (failed to download openapi: proto: cannot parse invalid wire-format data).
- **Kind Config Error:** Syntax error in kind-config.yaml at line 11: yaml: line 11: could not find expected ':'.

2. Kind Tool Installation

The `kind` command was not found on the system, so it was installed via Ubuntu package manager:

```
$ sudo apt install kind
Fetching 3632 kB in 2s (1510 kB/s)
Selecting previously unselected package kind.
Setting up kind (0.30.0-1)...
```

IMAGE 2

Kind Cluster Creation & Kubernetes Resources Deployment

1. Kind Cluster Setup

After installing Kind, a local Kubernetes cluster was initialized:

- **Config Error:** `kind create cluster --config kind-config.yaml` failed due to missing or invalid syntax for `apiVersion` on line 10.
- **Successful Cluster Creation:** Running the standard `kind create cluster` command successfully initialized the cluster (Node Image: `kindest/node:v1.34.0`).

2. Cluster Status & Manifests Deployment

After verifying cluster status, all Kubernetes resource manifests were applied:

```
$ kubectl get nodes
NAME STATUS ROLES AGE VERSION
```

```
kind-control-plane Ready control-plane 85s v1.34.0
```

```
$ kubectl apply -f .  
namespace/skillpulse created  
secret/skillpulse-db created  
configmap/mysql-init created  
service/mysql created  
statefulset.apps/mysql created
```

IMAGE 3

Pod Management, Port Forwarding & Debugging

1. Pod Deletion & Verification

Previous pod instances were deleted and verified:

- `kubectl delete pod backend-8476bcf67f-hx4rs -n skillpulse`
- `kubectl delete pod frontend-7757c66447-tdqcu -n skillpulse`
- Syntax correction: `kubectl get podes` threw an error (corrected to `pods`).

2. Running Pods Status

```
$ kubectl get pods -n skillpulse  
NAME READY STATUS RESTARTS AGE  
backend-8476bcf67f-wqr9v 1/1 Running 0 3m31s  
frontend-7757c66447-28qrw 1/1 Running 0 32s  
mysql-0 1/1 Running 0 12m
```

3. Port Forwarding & Editing

- **Port Conflict:** Forwarding port `8080:8080` threw a `bind: address already in use` error.
- **Successful Port Forward:** Port forwarding succeeded by switching to port `8081:8080`:

```
$ kubectl port-forward svc/backend 8081:8080 -n skillpulse  
Forwarding from 127.0.0.1:8081 -> 8080
```

- Opened `nano 20-backend.yaml` to modify configuration parameters.